

CHE210: DISCUSSION SESSION 5

OCTOBER 1, 2019

NAME: _____

DEGREES OF FREEDOM ANALYSIS:

1. Draw a flow diagram and choose a system.
2. Choose components.
3. Label the flow diagram to show the components in each stream and any accumulation in the system. Write down any chemical reactions.
4. Count the *number of independent variables*:

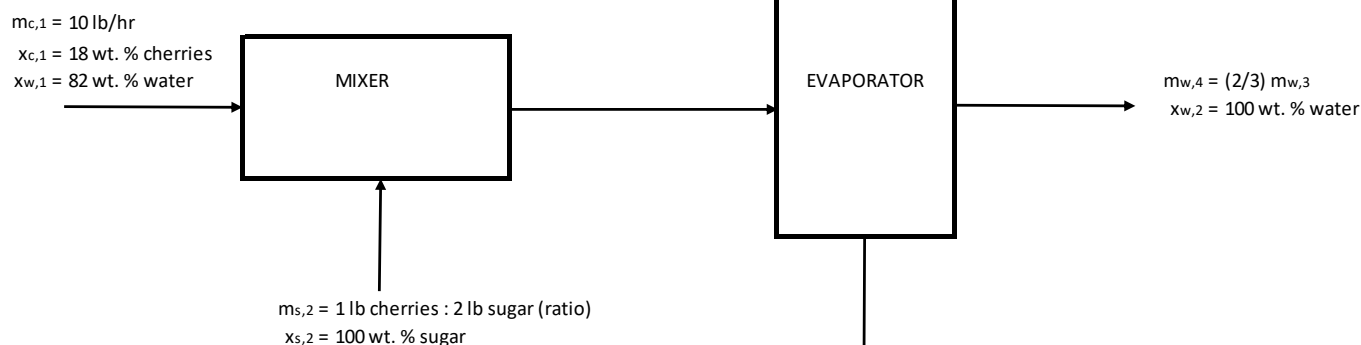
Do this by counting	
a. Count the number of independent stream variables.	The number of components in each stream, then adding them all up.
b. Count the number of independent system variables.	The number of independent chemical reactions, plus the number of components that accumulate (or are depleted) in the system.

5. Count the *number of independent equations*:

Comments	
a. Count the number of specified flows.	Usually there is only one flow specified, that serves as the basis.
b. Count the number of stream composition specifications.	If there are N components in a stream, there are at most $N - 1$ specified stream compositions. For a splitter, add $(N - 1)(N_s - 1)$, to the count, where N_s is the number of output streams.
c. Count the number of system performance specifications.	
d. Count the number of material balance equations.	If there are N components in a system, then there are N material balance equations.

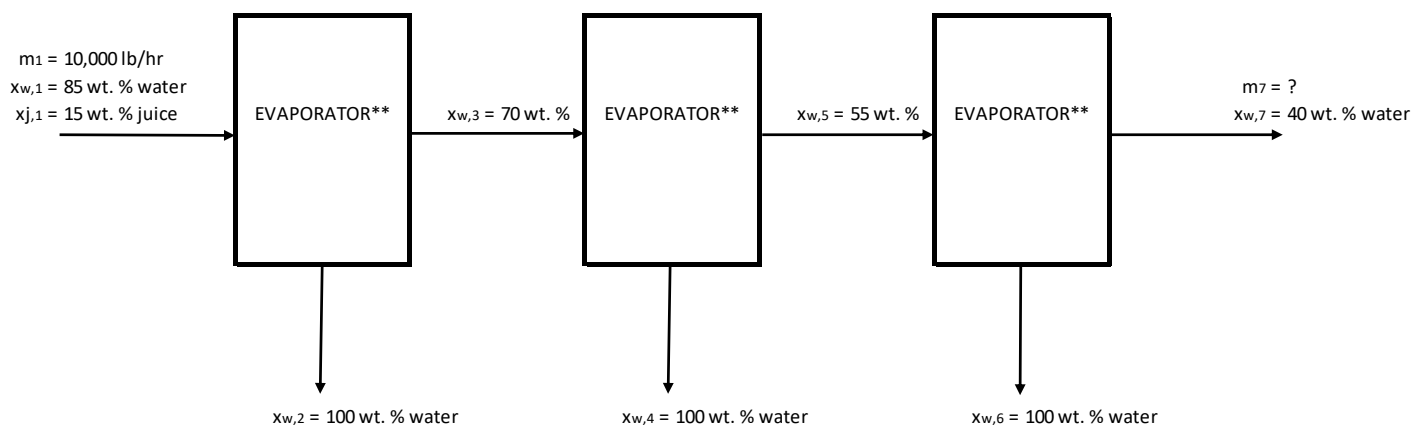
6. Calculate $DOF = \text{number of independent variables} - \text{number of independent equations}$.

P2.35 Cherry Jam Process



Unknowns:	Equations:
4a. _____	5a. _____
4b. _____	5b. _____
TOTAL _____	5c. _____
	5d. _____
	TOTAL _____

P2.48 Sugar Crystals from Sugar Cane



Unknowns:

Equations:

****NOTE:** Equal amounts of water removed from each evaporator.

4a. _____ 5a. _____

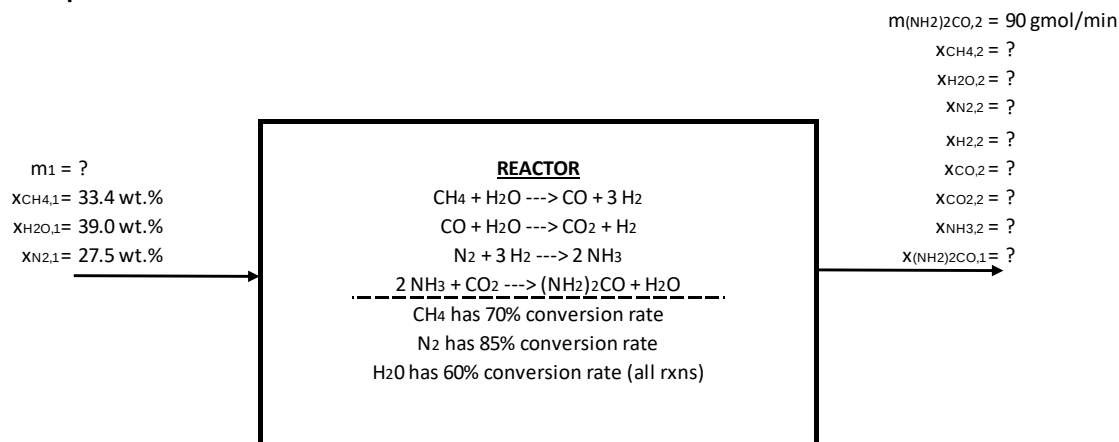
4b. _____ 5b. _____

TOTAL _____ 5c. _____

5d. _____

TOTAL _____

Modified Example: 3.6



Unknowns:

Equations:

4a. _____ 5a. _____

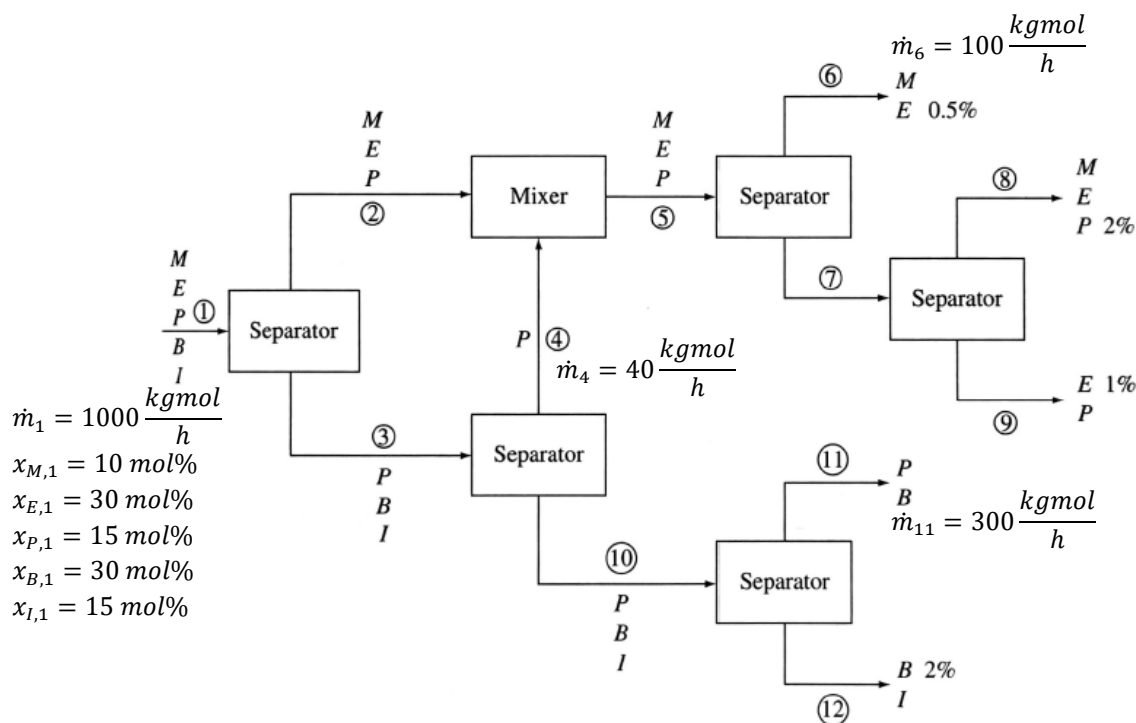
4b. _____ 5b. _____

TOTAL _____ 5c. _____

5d. _____

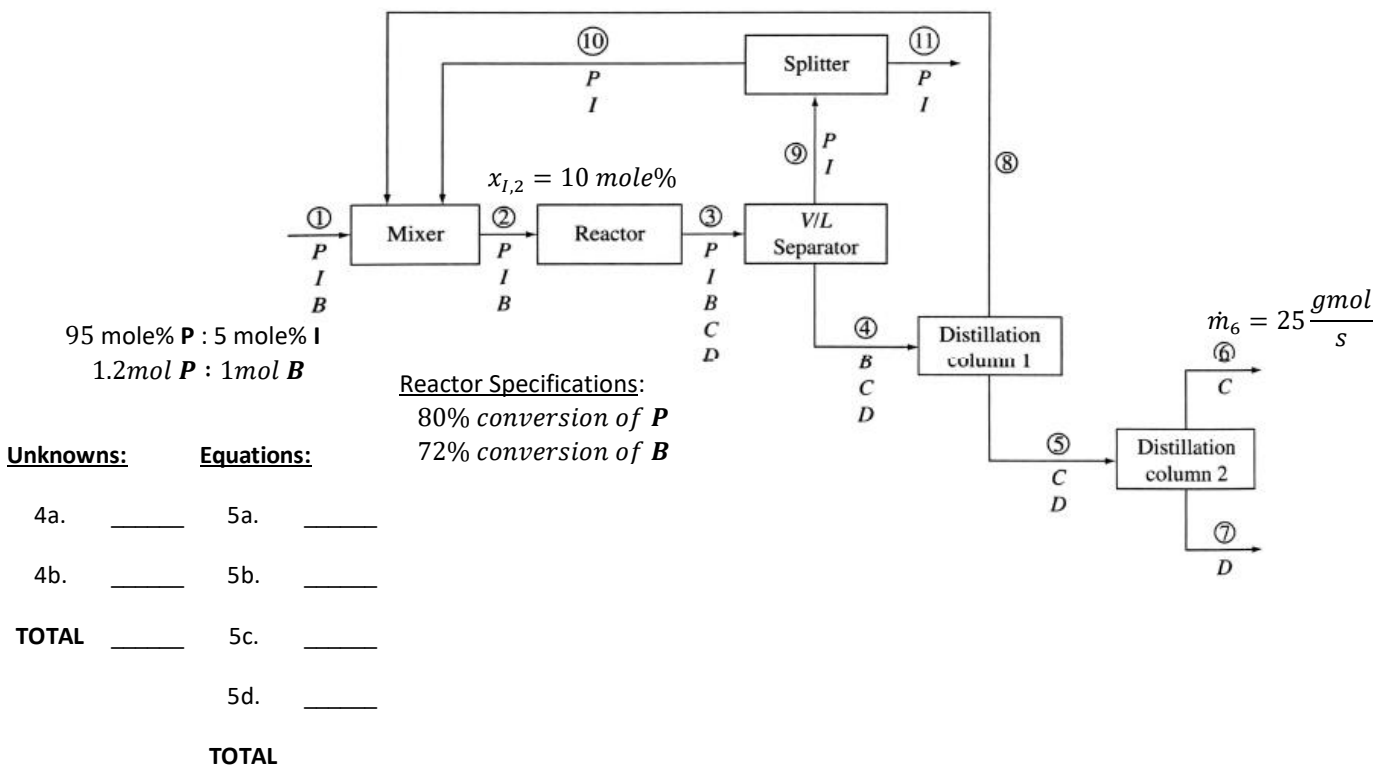
TOTAL _____

P2.70



Unknowns:	Equations:
4a. _____	5a. _____
4b. _____	5b. _____
TOTAL _____	5c. _____
	5d. _____
	TOTAL _____

P2.69



Unknowns:	Equations:
4a. _____	5a. _____
4b. _____	5b. _____
TOTAL _____	5c. _____
	5d. _____
	TOTAL _____

Combustion of a hydrocarbon fuel of unknown composition

A hydrocarbon gas is burned with air. The dry-basis product gas composition is 1.5 mole% CO, 6.0% CO₂, 8.2% O₂, and 84.3% N₂. There is no atomic oxygen in the fuel. Calculate the ratio of hydrogen to carbon in the fuel gas and speculate on what the fuel might be.

Homework Review: P3.24

Formaldehyde (HCHO) is produced by partial oxidation of methanol (CH₃OH). Several side reactions also occur, producing formic acid (HCOOH), CO, CO₂, and H₂O. Suppose 100 kgmol/h methanol and 21.05 kgmol/h O₂ are fed to a reactor, where 40% of the methanol and 95% of the O₂ is converted to products. What must the reactor outlet composition be?